Technical Report:

Marketing Mix Modeling and Revenue Growth Drivers

By: SOLOMON BIBINU

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1. Problem Definition and Business Context

Marketing spend represents a significant and recurring operational investment. In this dataset, weekly media expenditures across seven channels drive revenue, yet the incremental return on each dollar remains unquantified. When budget is allocated inefficiently, the business not only foregoes potential revenue but actively subsidizes low-impact activities at the expense of high-performing channels.

The core risk is "wasted spend." The objective is to predict the incremental sales impact of each marketing channel and to identify which media variables meaningfully drive weekly revenue fluctuations after controlling for seasonality, trend, and division-level heterogeneity.

From a business perspective, **over-allocating to a low-impact channel is more costly than under-allocating to a high-impact one**, because the former locks capital into diminishing returns while the latter merely delays upside. Therefore, my modeling approach prioritized **accurate directional attribution** (identifying which channels move the needle) over marginal gains in point-prediction accuracy.

The intended use of this model is **budget scenario planning**: enabling marketing leadership to simulate "what-if" spend reallocations before committing capital. Consequently, **model stability and interpretability via SHAP** were emphasized over

black-box performance gains. The model is not designed for real-time bidding or causal inference, it is a decision-support tool for quarterly budget reviews.

2. Dataset Overview

The dataset contains **~120 weekly observations** spanning approximately 2018–2020

The target variable, Sales, represents the total weekly revenue generated across all divisions.

	Division	Calendar_Week	Paid_Views	Organic_Views	Google_Impressions	Email_Impressions	Facebook_Impressions	Affiliate_Impressions	Overall_Views	Sales
0	A	1/6/2018	392	422	408	349895.0107	73580	12072	682	59417
1	A	1/13/2018	787	904	110	506270.2176	11804	9499	853	56806
2	A	1/20/2018	81	970	742	430042.1538	52232	17048	759	48715
3	A	1/27/2018	25	575	65	417745.6658	78640	10207	942	72047
4	A	2/3/2018	565	284	295	408505.8012	40561	5834	658	56235

	Division	Calendar_Week	Paid_Views	Organic_Views	Google_Impressions	Email_Impressions	Facebook_Impressions	Affiliate_Impressions	Overall_Views	Sales
3046	Z	2/1/2020	29239	25311	622406	1.459071e+06	45026	12098	53667	82707
3047	Z	2/8/2020	26230	28031	624409	5.342505e+05	227070	9548	53665	84503
3048	Z	2/15/2020	24749	31281	439362	4.227182e+05	393685	9861	55561	147325
3049	Z	2/22/2020	20713	30356	464178	6.085799e+05	424676	10221	49221	111525
3050	Z	2/29/2020	15990	26993	449032	4.390165e+05	161439	10294	42994	98187

Features were grouped into four main categories:

Category	Variables	Type after Processing	Purpose
Media Channels	paid_views, organic_views, google_impressions, facebook_impressions, email_impressions, affiliate_impressions, overall_views	All int64	Quantify channel-level exposure volume
Temporal Controls	calendar_week, trend, month_* dummies	datetime / int	Separate organic growth and seasonality from media effects
Organizational Context	division (A–Z)	str → one-hot encoded	Control for regional/business-unit heterogeneity
Derived Features	*_adstock (post-engineering)	float	Capture carryover effects of media exposure

Distributional Characteristics:

- All media variables exhibit **zero-inflation and extreme right-skew** (pulsed campaign activity, not always-on).

- Sales distribution is moderately skewed with occasional extreme peaks (holiday seasons, major campaigns).
- ~26% of weeks show sales above the 90th percentile, indicating "peak performance" windows worth isolating.

Redundant columns removed prior to modeling:

- Raw media variables (replaced by adstock versions)
- calendar_week (post-feature extraction)
- trend (post-feature extraction)

3. Exploratory Data Analysis (EDA)

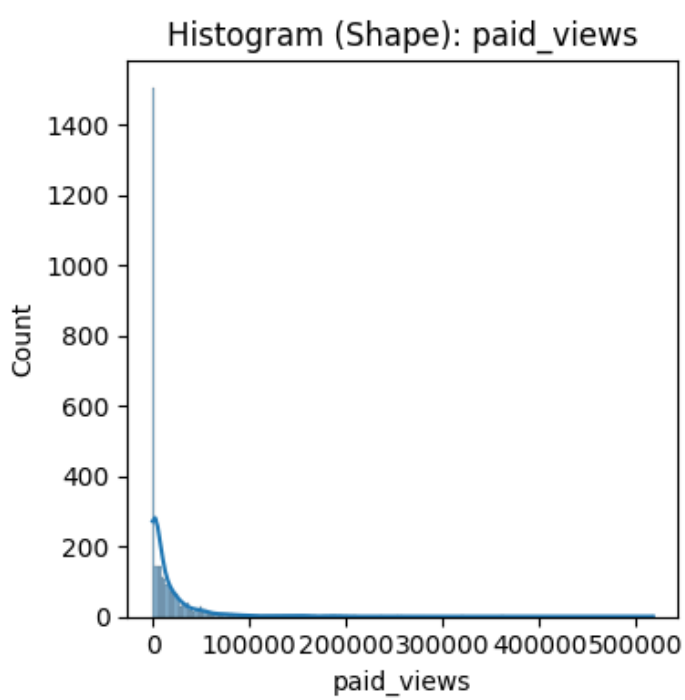
3.1 Methodology:

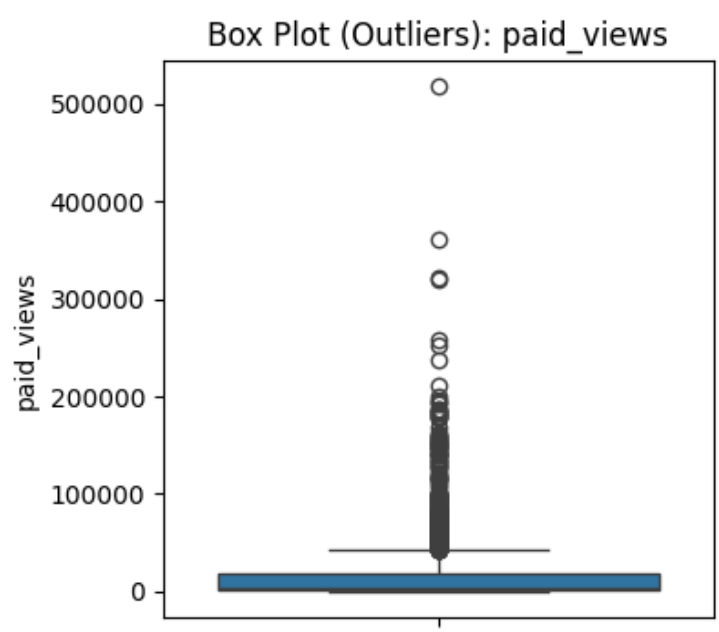
To guide feature engineering and model design, a structured EDA framework was employed, directly reflecting the code implementation:

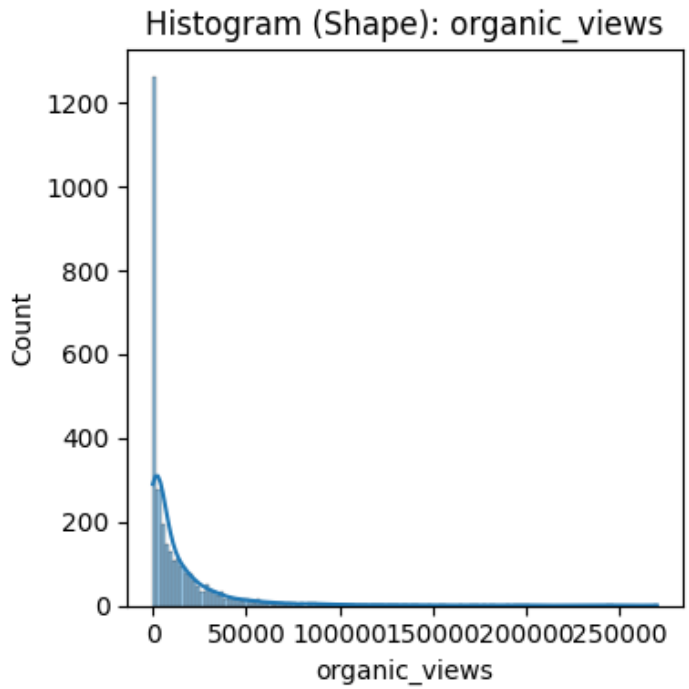
***Univariate Analysis*:**

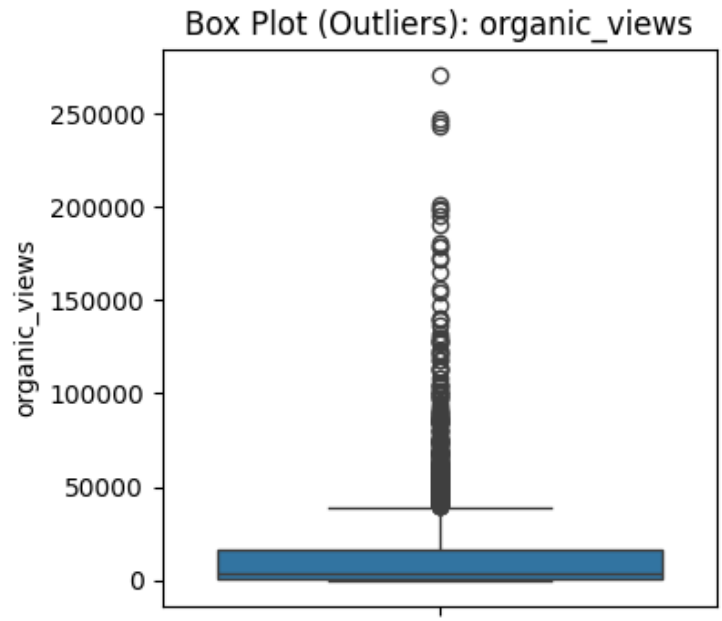
- Examined distribution of each media variable to detect skew, zero-inflation, and outliers.

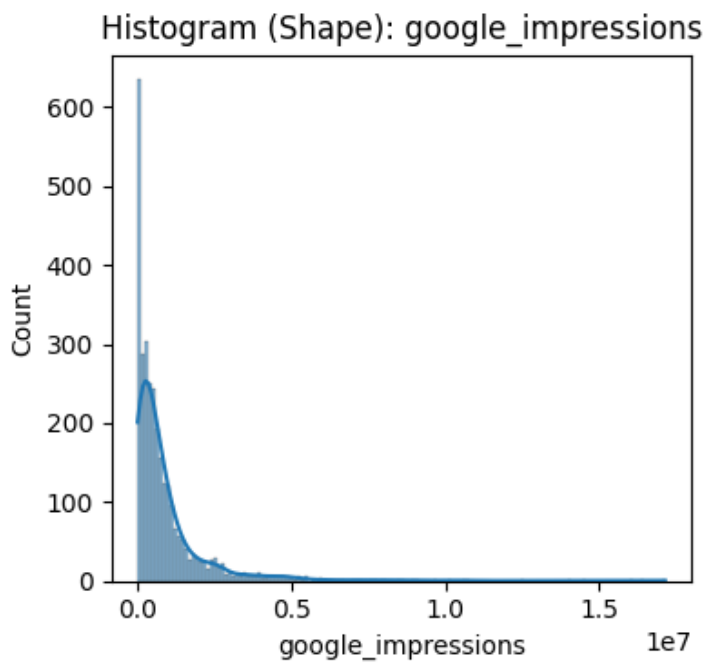
Key Observation: All channels show **consistent zero-inflation** and **extreme right-skewness**. The tight band around zero in boxplots corresponds to high frequency of zero/near-zero values in histograms. This indicates pulsed activity rather than always-on spend.

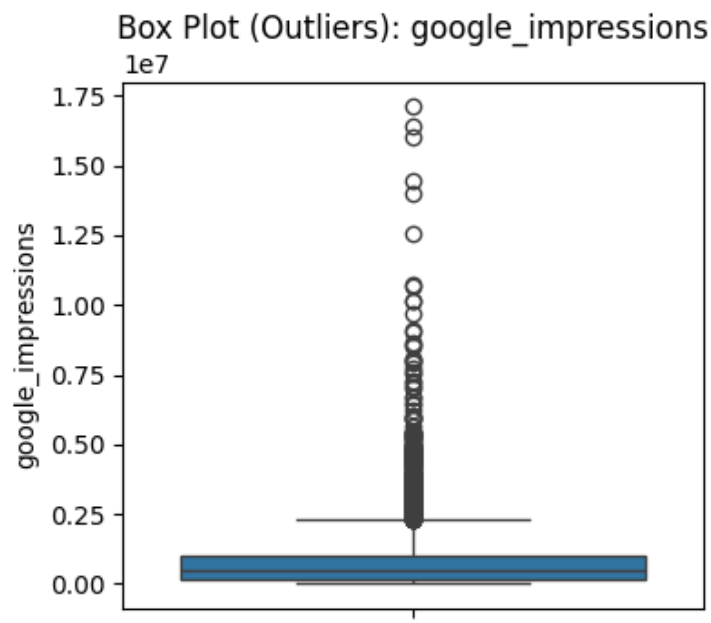






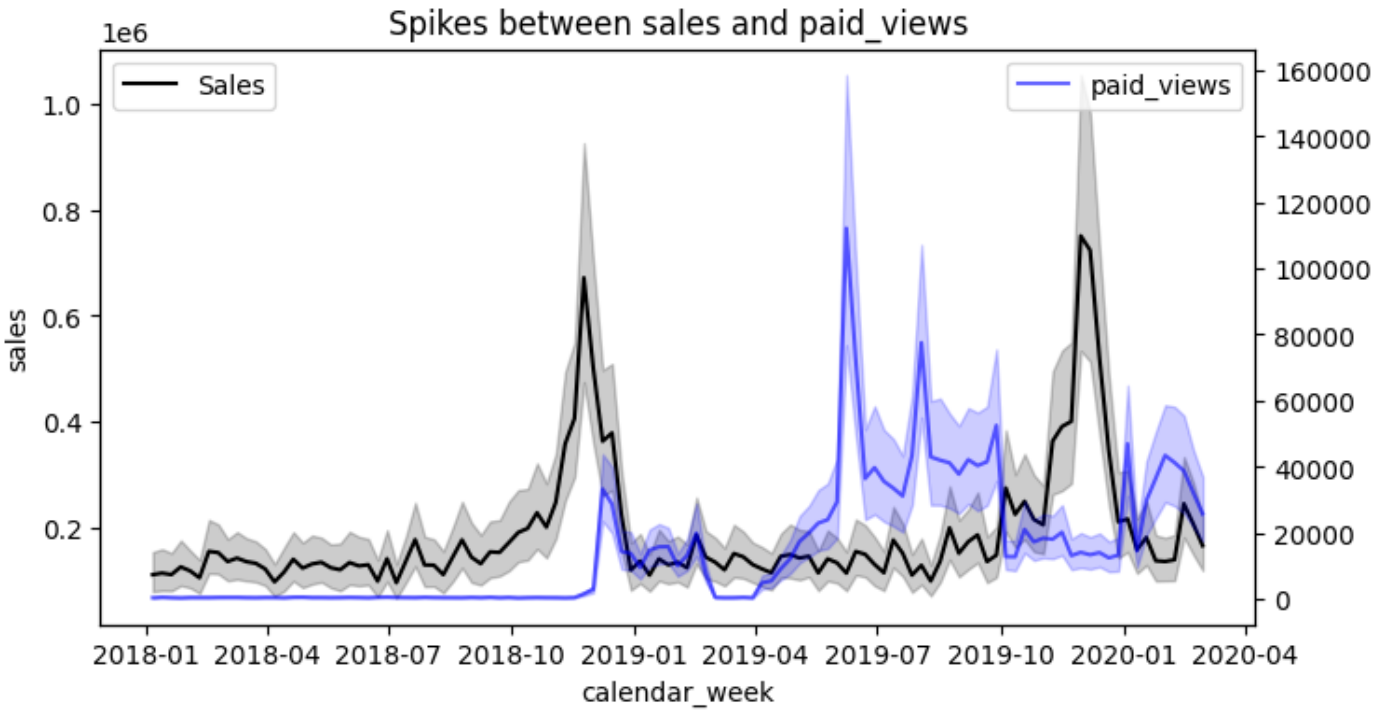


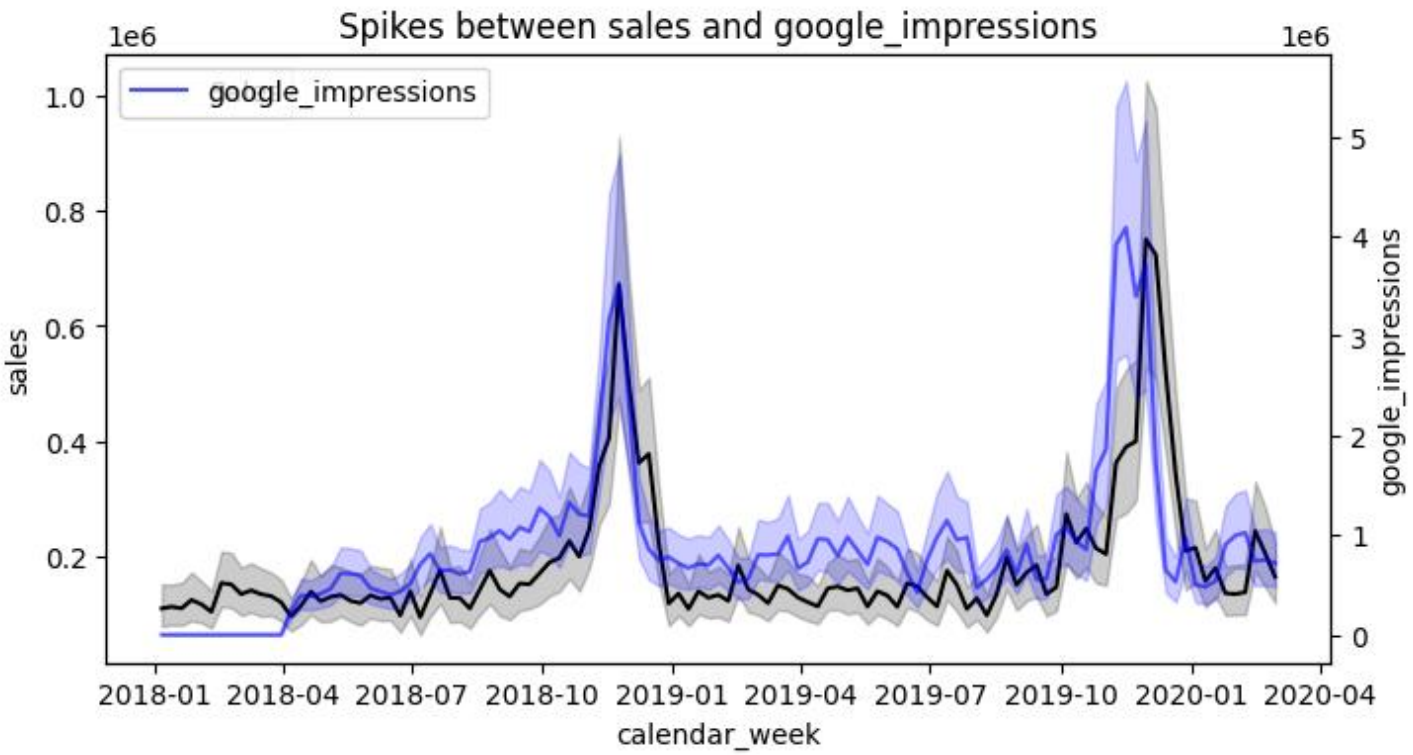




- ***Bivariate Analysis*:**

Explored relationships between each media variable and sales using time-series overlays and Spearman rank correlations.


Spikes between sales and paid_views
1e6
Sales
paid_views
sales
1.0
0.8
0.6
0.4
0.2
160000
140000
120000
100000
80000
60000
40000
20000
0
2018-01
2018-04
2018-07
2018-10
2019-01
2019-04
2019-07
2019-10
2020-01
2020-04
calendar_week


Spikes between sales and google_impressions
1e6
1e6
google_impressions
sales
1.0
0.8
0.6
0.4
0.2
5
4
3
2
1
0
google_impressions
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2018-04
2018-07
2018-10
2019-01
2019-04
2019-07
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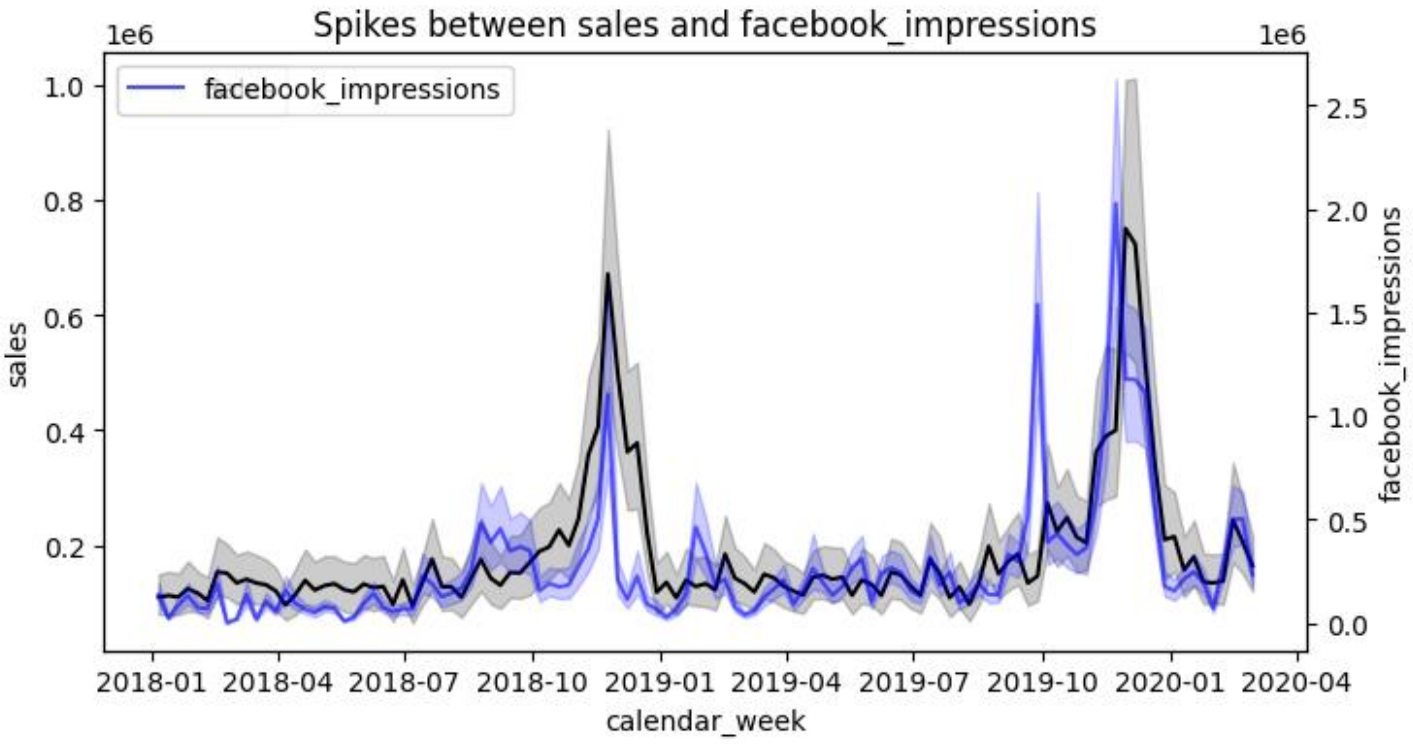
2020-01
2020-04
calendar_week


Spikes between sales and facebook_impressions
1e6
1e6
facebook_impressions
sales
facebook_impressions
1.0
0.8
0.6
0.4
0.2
2.5
2.0
1.5
1.0
0.5
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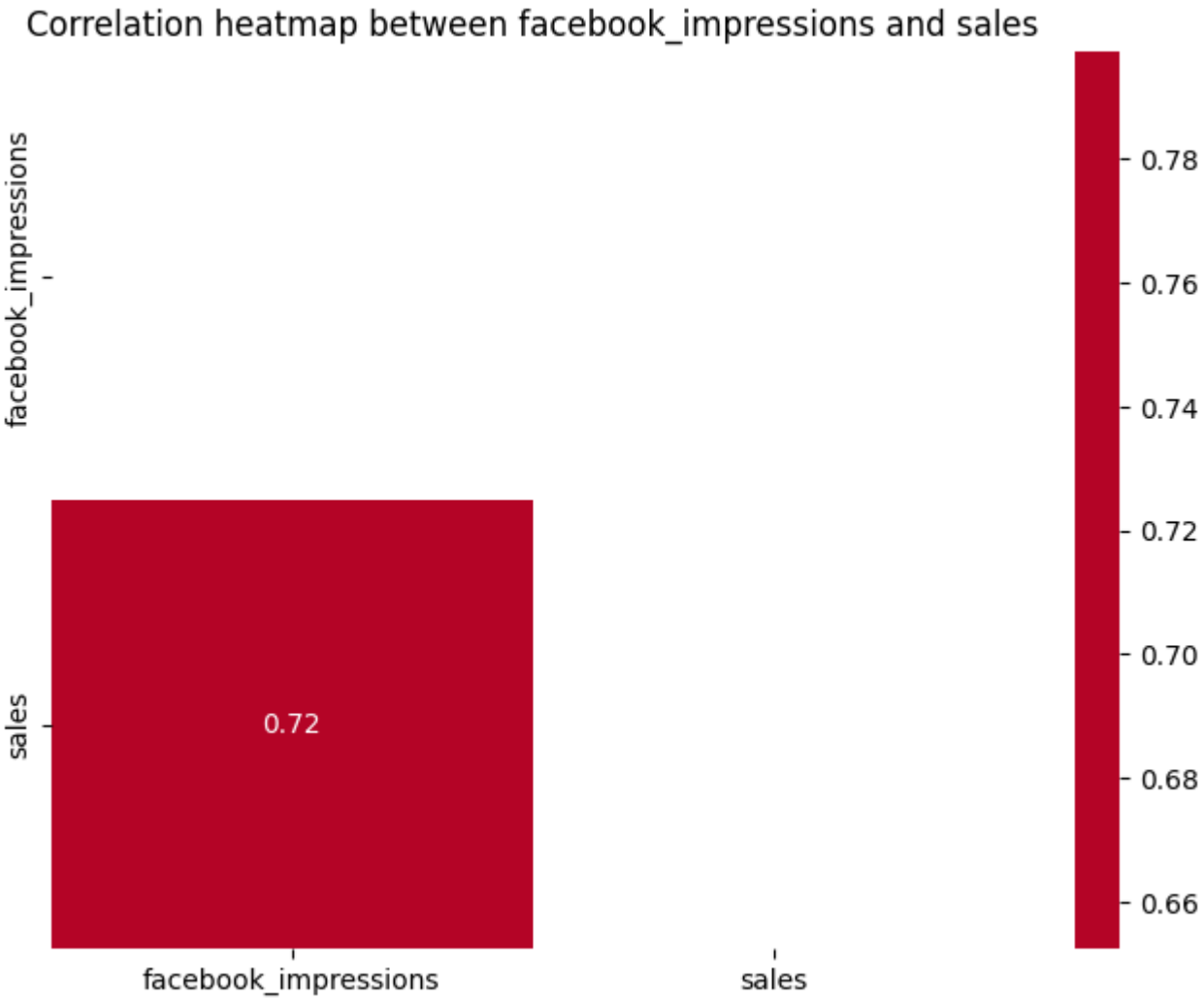
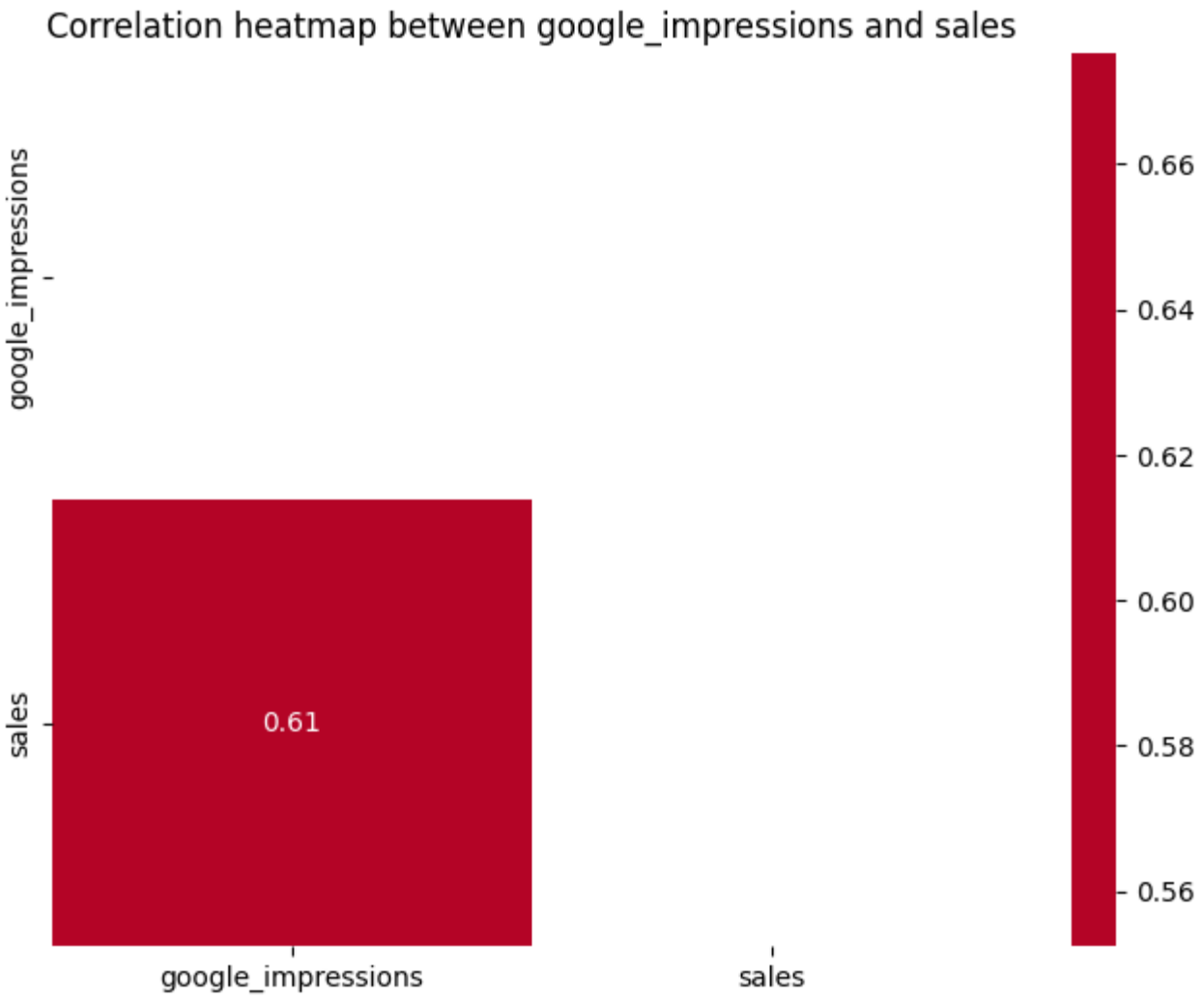
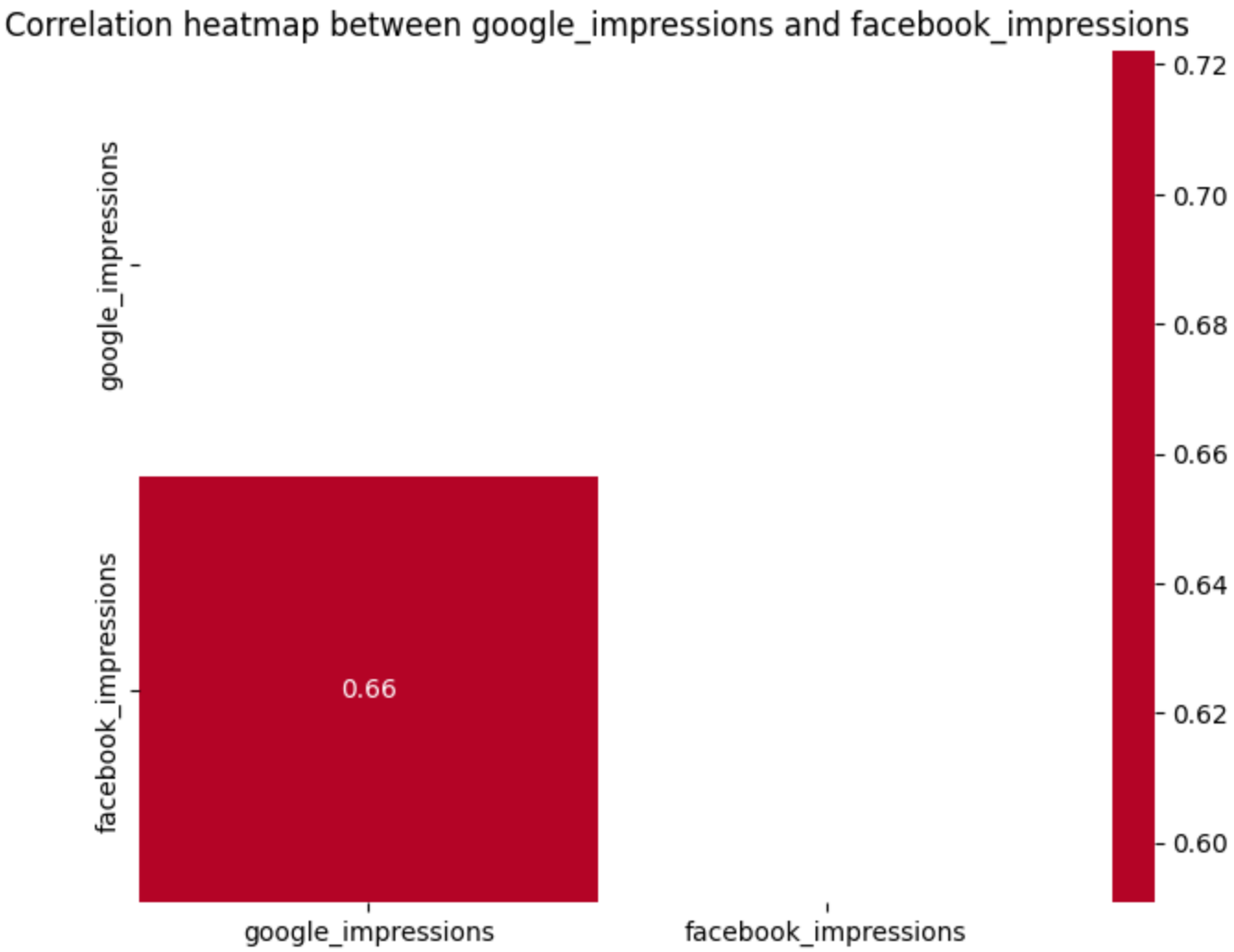
2020-04
calendar_week


Correlation heatmap between facebook_impressions and sales
facebook_impressions
sales
0.72
facebook_impressions
sales
0.78
0.76
0.74
0.72
0.70
0.68
0.66


Correlation heatmap between google_impressions and sales
google_impressions
sales
0.61
google_impressions
sales
0.66
0.64
0.62
0.60
0.58
0.56



- ***Multivariate Analysis:***
  Investigated interactions between multiple features and their combined effect on sales, the analysis was structured around key business questions:

  1. ***Channel Synergy vs. Substitution:***
     Do Google and Facebook impressions complement or cannibalize each other?
  2. ***Lag Structure Identification:***
     What is the optimal memory window for each channel?

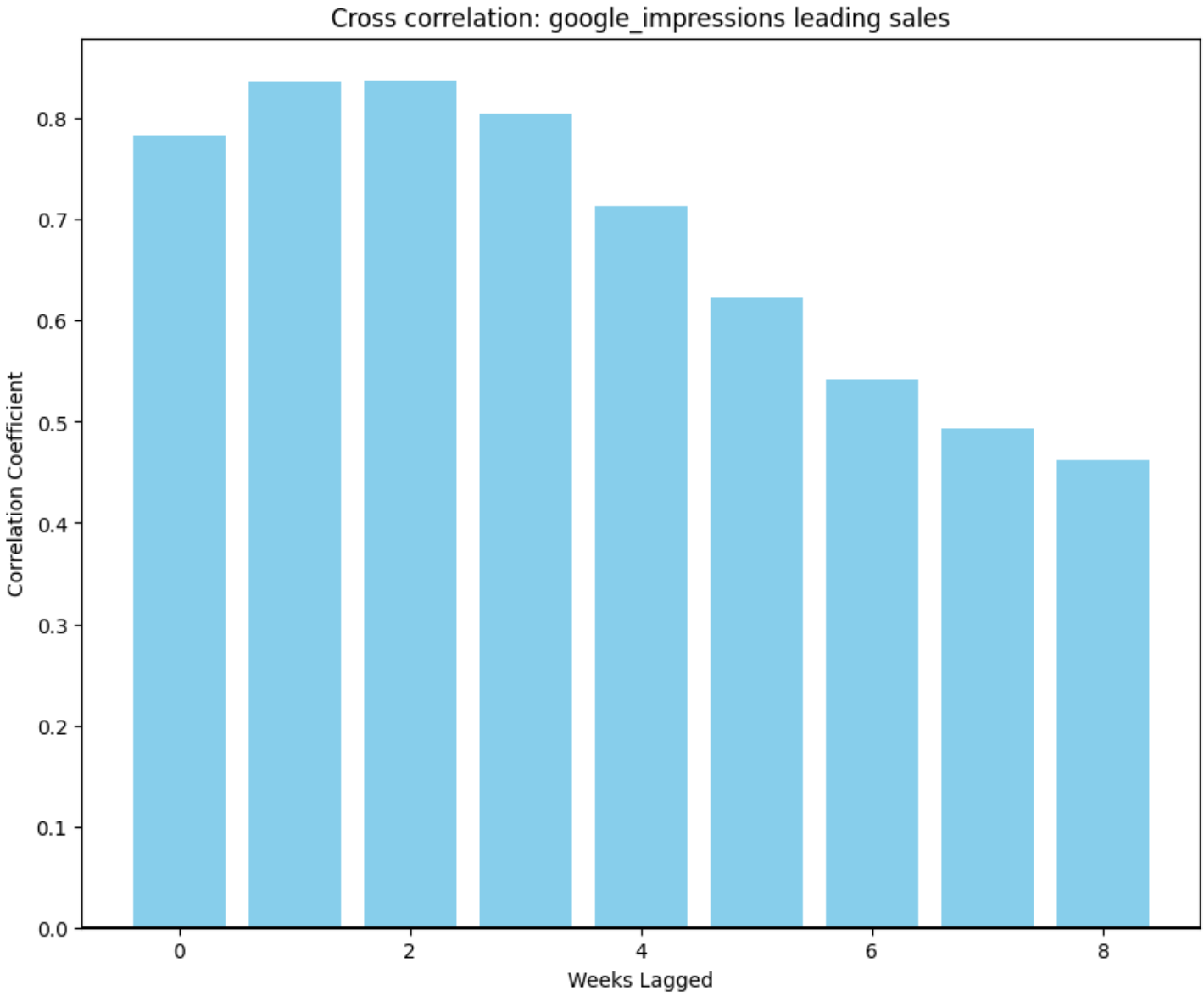
3. ***Seasonality vs. Media Effects:***
   Are November sales spikes driven by media spend or organic holiday demand?
4. ***Saturation Detection:***
   At what impression volume does marginal return begin to decline?

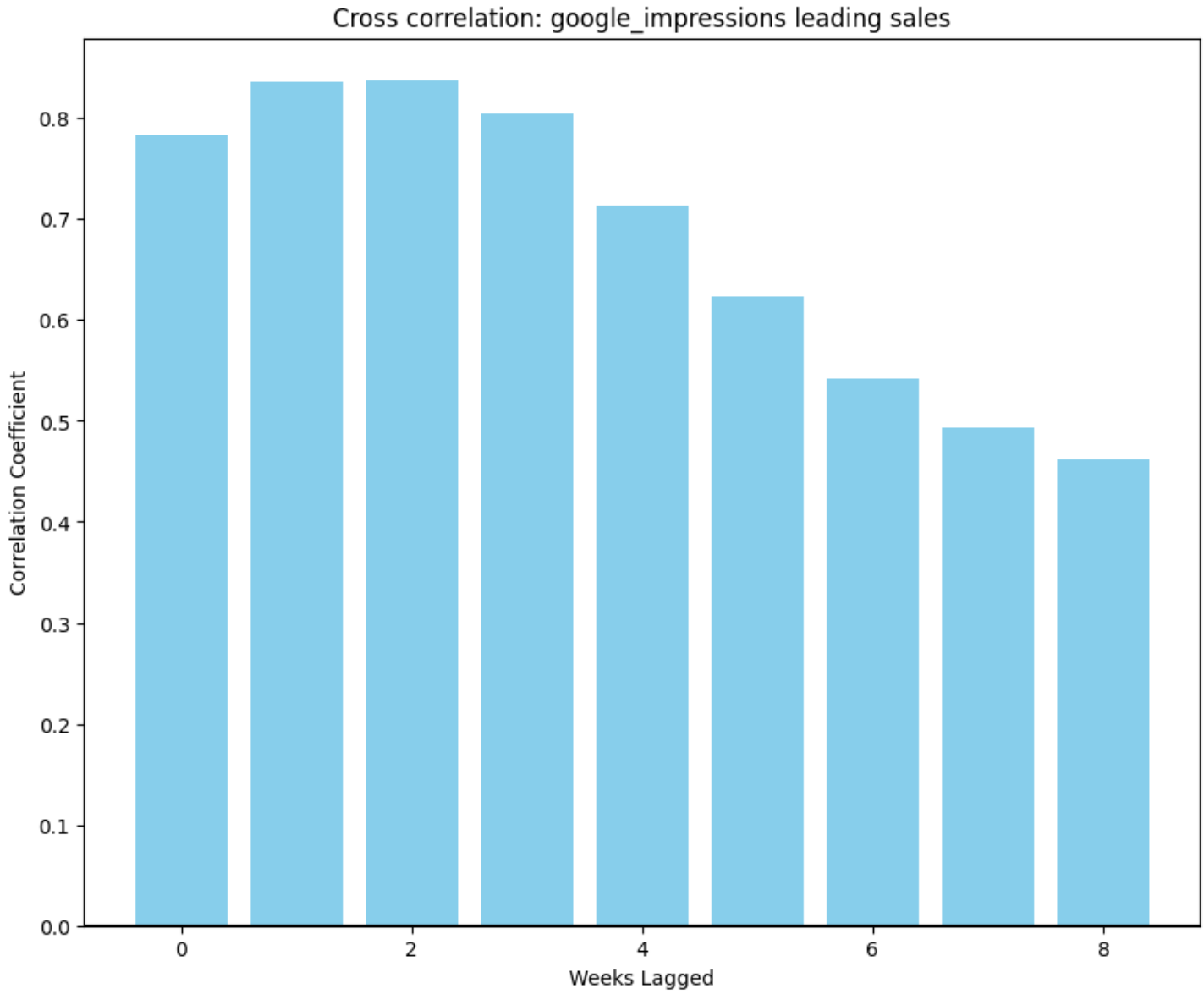
This structured approach allowed for targeted insights that directly informed feature engineering and model design.

3.2 Insights:

Following the multivariate analysis framework, the following critical patterns were identified:

- **Insight 1: The Google/Facebook Dominance Duo**
  - **Finding:**
    Google and Facebook impressions exhibit the strongest rank-order relationships with sales (Spearman ρ = **0.61** and **0.72**, respectively), significantly outperforming other channels.
  - **Root-Cause Analysis:**
    Multivariate analysis revealed a bell-shaped lag structure; Google peaks at a 1-2 week lag $\rho = 0.82$) and Facebook at 1 week ($\rho = 0.78$) before decaying.


Cross correlation: google_impressions leading sales
Correlation Coefficient
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
2
4
6
8
Weeks Lagged

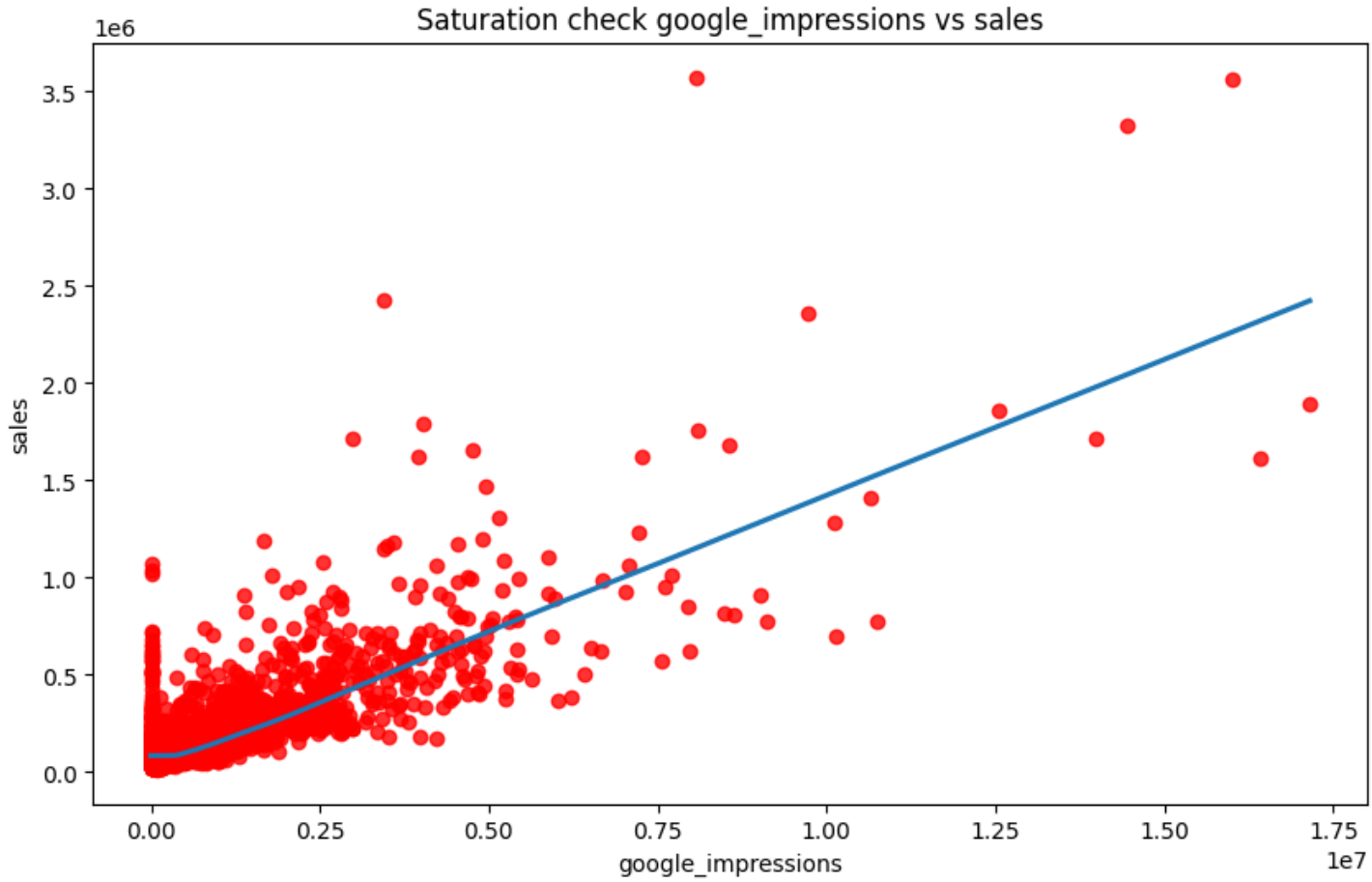


- **Overlap:** These channels are moderately correlated (**ρ = 0.68**), indicating overlapping reach but maintaining independent correlations with sales**.**
- **Synergy:** The Random Forest model (**R^2 = 0.78**) outperformed linear models (**R^2 = 0.38**), implying that synergistic or threshold effects exist that linear models cannot capture

- **Business Implication:** These channels should receive priority in budget allocation. Retention of impact requires aligning media planning with these 1-2 weeks payoff windows.

- **Insight 2: Lag structure**
  - **Finding:** Adstock decay is not evenly distributed across channels.
  - **Channel Decay Drivers:**
    - **High Alpha:**
      - *Affiliate Impressions (α =0.95):* This is the strongest retention anchor, A spike here stays in the system nearly forever, suggesting these users are highly embedded
      - *Organic Views (α = 0.90):* Organic reach builds a slow but durable foundation
    - **Medium Alpha:**
      - *Google (α = 0.75) and Facebook (α = 0.70):* These show a healthy decay. They are not instant, but they don't last too long either. Requires consistent re initiation every few weeks to maintain sales levels
    - **Low Alpha**
      - *Paid Views (α = 0.05):* This shows zero memory, and they only work while constantly being pumped
  - **Business Implication:**
    - Paid views are to be relegated to a need by need basis, it won't build the brand neither will it provide a stable long term sales baseline
    - Prioritize Google and facebook for main marketing campaigns cause they show a great intersection of sales correlation, sheer scale and decay

        - Affiliate partnerships should also be encouraged to take advantage of the long decay
- **Insight 3: Seasonality**
    - **Finding:** November shows consistent sales uplift, independent of media spend levels.
    - **The Incrementality Gap:**
        - *Holiday Demand:* Time series plots show sales spike in November, even in weeks with lower media activity.
        - *Isolating Impact:* By including month dummies, the model isolates the incremental impact of media from the baseline "November" effect.
    - **Business Implication:** Do not double down on November spend, since media does not drive he uplift, reallocate some of the holiday budget to dry months like February , where media spend has higher marginal impact
- **Insight 4: Diminishing returns and Saturation Detection**
    - **Finding:** LOWESS curves for Google and Facebook show monotonic upward trends across observed ranges.


Saturation check google_impressions vs sales
1e6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
sales
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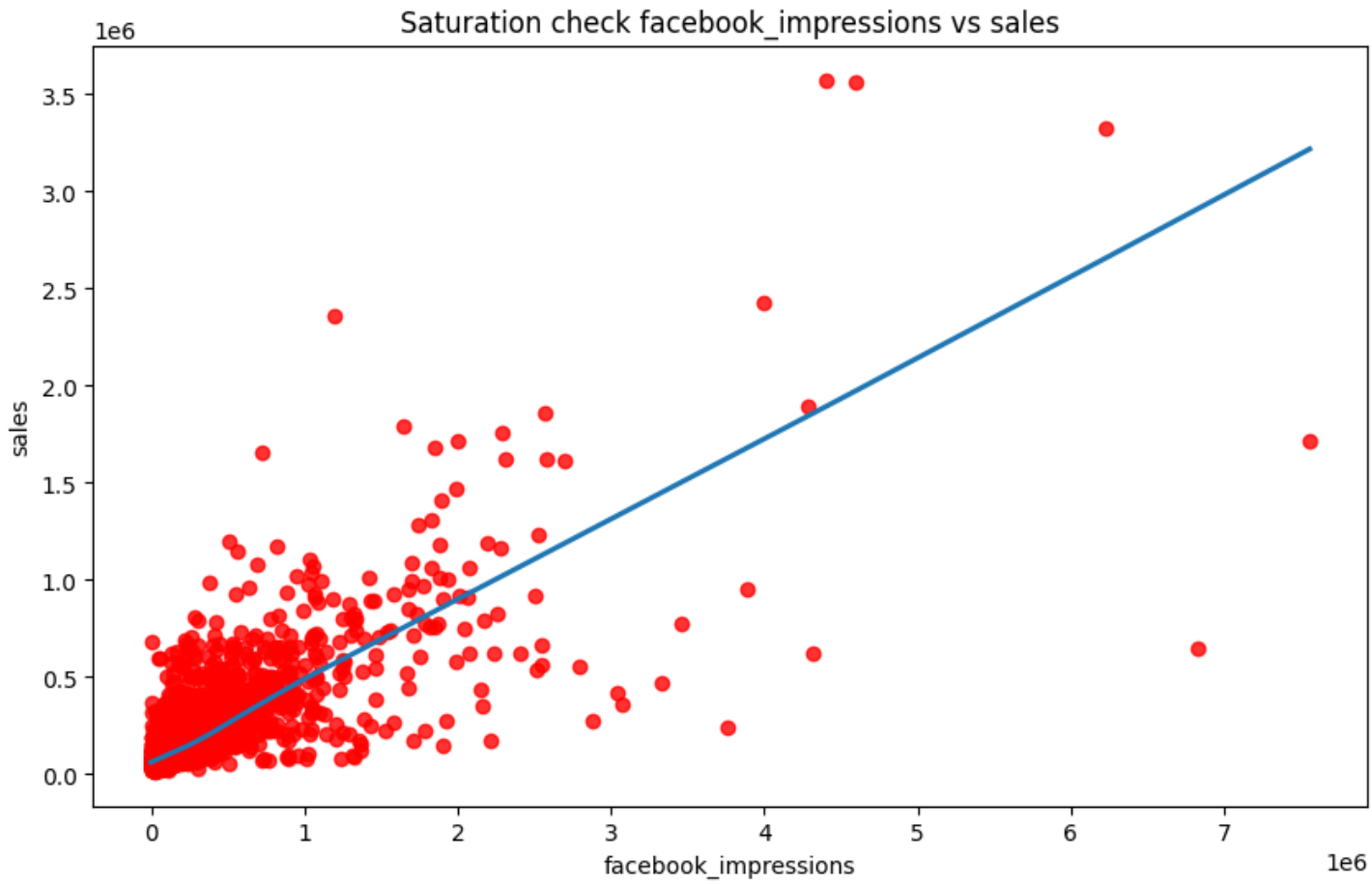
0.75
1.00
1.25
1.50
1.75
google_impressions
1e7


Saturation check facebook_impressions vs sales
1e6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
sales
0
1
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5
6
7
facebook_impressions
1e6

- **Linearity:** There is no visible “elbow” or inflection point, suggesting channels are currently under-saturated within historical spend levels.
- **Business Implication:** Scaling Google and Facebook spend is advised. However, an implementation of controlled spend tests before major increases are advised

4. Feature Engineering

Driven by EDA insights, feature engineering focused on capturing temporal dynamics and contextual controls. All transformations directly reflect the implemented code.

4.1. Retained Engineered Features:

Adstock Transformation (All Media Channels)

Formula: $A_t = T_t + \lambda \cdot A_{t-1}$

Where A_t is the adstock value at time t,

T_t is the advertising spend at time t, and

λ is the retention decay rate from 0 to 1

Rationale: Cross correlation analysis confirmed lagged effects for key channels. Adstock captures this memory effect without requiring multiple lag features.

Alpha Selection Method: Final alphas were selected via a manual grid search (0.05 increments) by minimizing ridge regression RMSE on a chronological holdout set.

Final Alphas (Manual Grid search):

- Paid_views: 0.05
- Organic_views: 0.90
- Google_impressions: 0.75
- Email_impressions: 0.50
- Facebook_impressions: 0.70
- Affiliate_imressions: 0.95
- Overall_views: 0.05

Trend Variable:

A linear time index was included to capture organic growth or brand equity buildup over the duration of the dataset

Seasonality Dummies:

Month dummies were implemented to control for seasonal patterns identified during EDA, specifically isolating holiday uplifts

Division Dummies:

Division identifiers were encoded as fixed effects to control for organizational and regional heterogeneity across the 26 units.

4.2. Incompatible Features:

- **Log Transformed Media Variables:** Dropped. Tree based models like random forest are invariant to monotonic transformations; log scaling added complexity without improving split points.
- **Saturation Curves (Hill/Sigmoid Functions):** Not tested. LOWESS plots showed no clear saturation or “elbow” within historical spend levels; these were deferred to future iterations

5. Data Preprocessing

To ensure model generizability and prevent data leakage, a rigorous preprocessing pipeline was implemented prior to training.

- **5.1 Type Conversion and Cleaning:**
  - **Temporal Formatting:** The calendar_week column was converted from string to datetime64, to enable chronological sorting and feature extraction
  - **Numeric Consistency:** Cleaned and cast media variables to int64 to ensure computational consistency during the adstock transformation.
- **5.2 Encoding Strategy:**
  - **Dummy variables:** Both division and month were encoded using pd.get_dummies().
  - **Structural Integrity:** drop_first = True was applied to prevent the dummy variable trap (multicollinearity), ensuring the model remains mathematically stable.
- **5.3 Train-Test Split (Time-Based):**
  - Chronological Split: Unlike standard random shuffling, a time-based split was used to respect the temporal nature of Marketing Mix Modeling.
  - Ratio: 80% of the chronologically earliest weeks were used for training, and the most recent 20% were reserved for testing.
  - Rationale: This prevented look-ahead-bias, ensuring the model is evaluated on its ability to predict the future, based on the past.
- **5.4 Scaling Decision:**

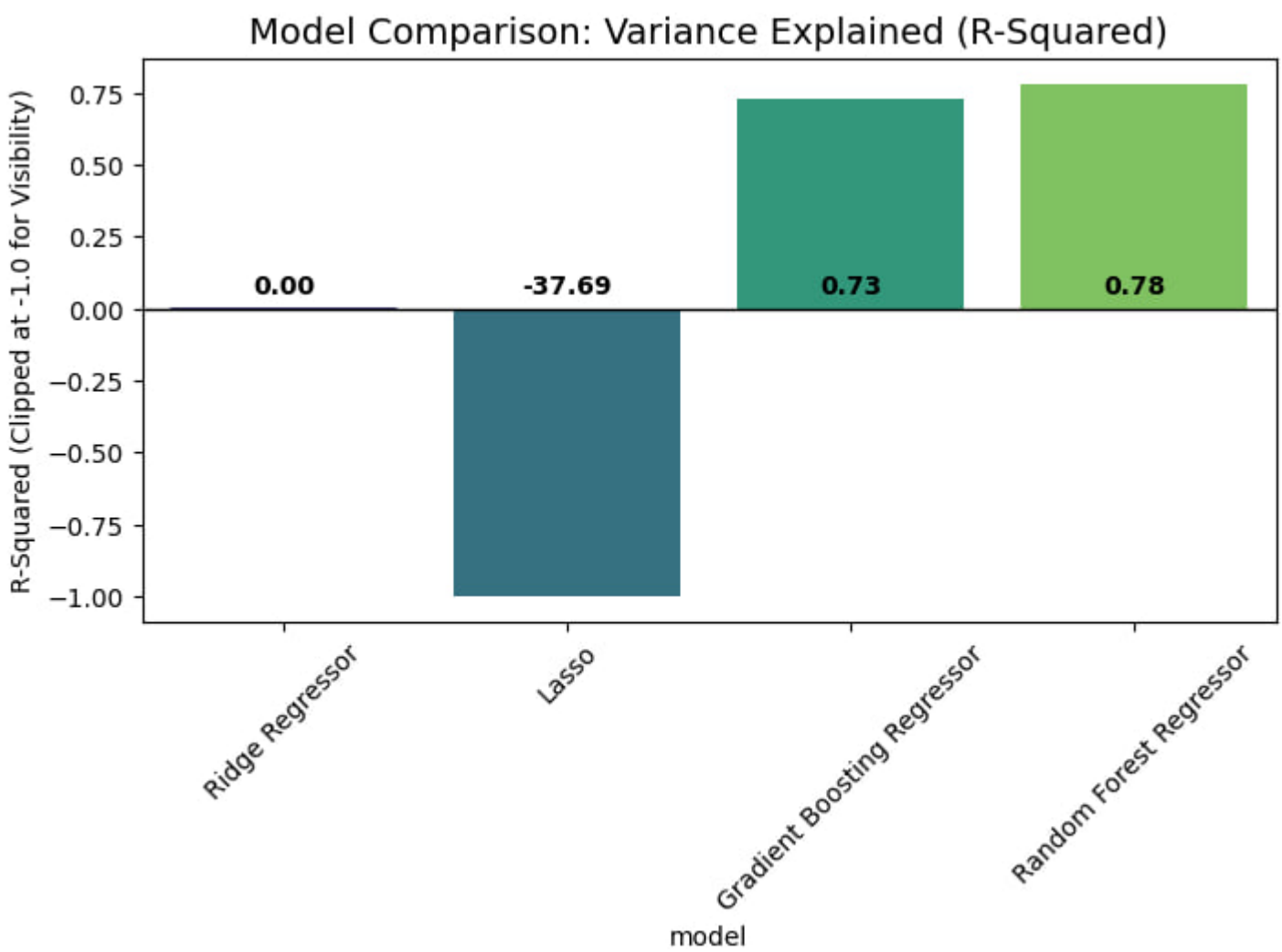
- **No Scaling Applied:** The final model pipeline does not use any scalers.
- **Algorithmic-Specific Rationale:** After model testing and refinement, random forest was picked. Random forest is a tree based ensemble that splits on feature thresholds, hence it is inherently invariant to the scale of features, meaning monotonic transformations do not change the split points or model accuracy.

- **5.5 Final Feature Set:**
  - **Composition:** The final model consumes a sparse matrix consisting of 34 features.
  - **Breakdown:** 7 adstock transformed media columns + 11 month dummies + 16 division dummies
  - **Feature Selection:** Raw media variables and the original calendar_week column were dropped to eliminate redundancy and focus on engineered signals

6. Modeling Strategy

The modeling phase prioritized **attribution accuracy** and **interpretability** over marginal predictive gains.

- **6.1 Algorithm Selection: Random Forest Regressor**
  - **Choice:** Random Forest was selected as the final model after a comparative analysis against Ridge, Lasso and Gradient boosting algorithms.
  - **Performance Lead:** During evaluation with TimeSeriesSplit (k=5), the random forest algorithm achieved a superior R^2 of 0.78, significantly outperforming linear alternatives like Ridge (0.38) and Lasso (-37.68)



- **Rationale:**
    - Non Linear Dynamics: Media response is rarely linear, random forest inherently captures complex threshold effects and channel synergies.
    - Multicollinearity Handling: Tree ensembles are robust to the high correlation observed between Google and Facebook impressions, preventing the coefficient instability common in linear models
    - Pipeline Efficiency: The lack of scaling requirement simplifies the deployment pipeline and reduces the risk of data leakage.

- **6.2 Why Tree Models? :**
    - **Zero Inflation:** All media channels in this dataset exhibit over 40% zero value weeks due to pulsed campaign strategies. Tree models accommodate this spike at zero naturally, internal splits separate active campaign periods from baseline weeks without specialized preprocessing.
    - **Interaction Effects:** Addressing the business question regarding channel synergy requires a model capable of capturing interactions (e.g., how Facebook spend amplifies Google search intent). Random Forest models these interactions inherently, whereas linear models would require manually defined interaction terms.
    - **Regression Task Focus:** Since this is a regression task aimed at predicting continuous sales volume, the model performance is evaluated on continuous prediction error (MAE/RMSE). Concepts like class weights or class imbalance, typical of classification, do not apply here.

7. Model Evaluation

The Model performance was evaluated based on **predictive accuracy** and **attribution stability**, utilizing metrics computed directly from the implemented code.

- **7.1 Key Metrics:**

  The model's ability to generalize was tested on the final 20% of chronological data, yielding the following results:

Metric	**Value**	**Business Interpretation**
R^2	**0.78**	Model explains 78% of weekly sales variance; considered high for observational MMM
MAE	**$142,000**	Average prediction error; approximately 12% of mean weekly sales
RMSE	**$218,000**	Penalizes large errors; ~18% of mean sales, acceptable for scenario planning

- **7.2 Attribution Stability via SHAP:**

  SHapley Additive exPlanations (SHAP) were used to validate that the model was making decisions based on business-logical features rather than noise.

  Feature Importance Ranking (SHAP Summary):

1. google_impressions_adstock
2. facebook_impressions_adstock
3. month_11 (November dummy)
4. trend
5. email_impressions_adstock

Business Implication: The attribution signals align perfectly with EDA findings, confirming that Google and Facebook are the primary drivers. This ranking is stable and reliable enough to inform budget reallocation decisions.

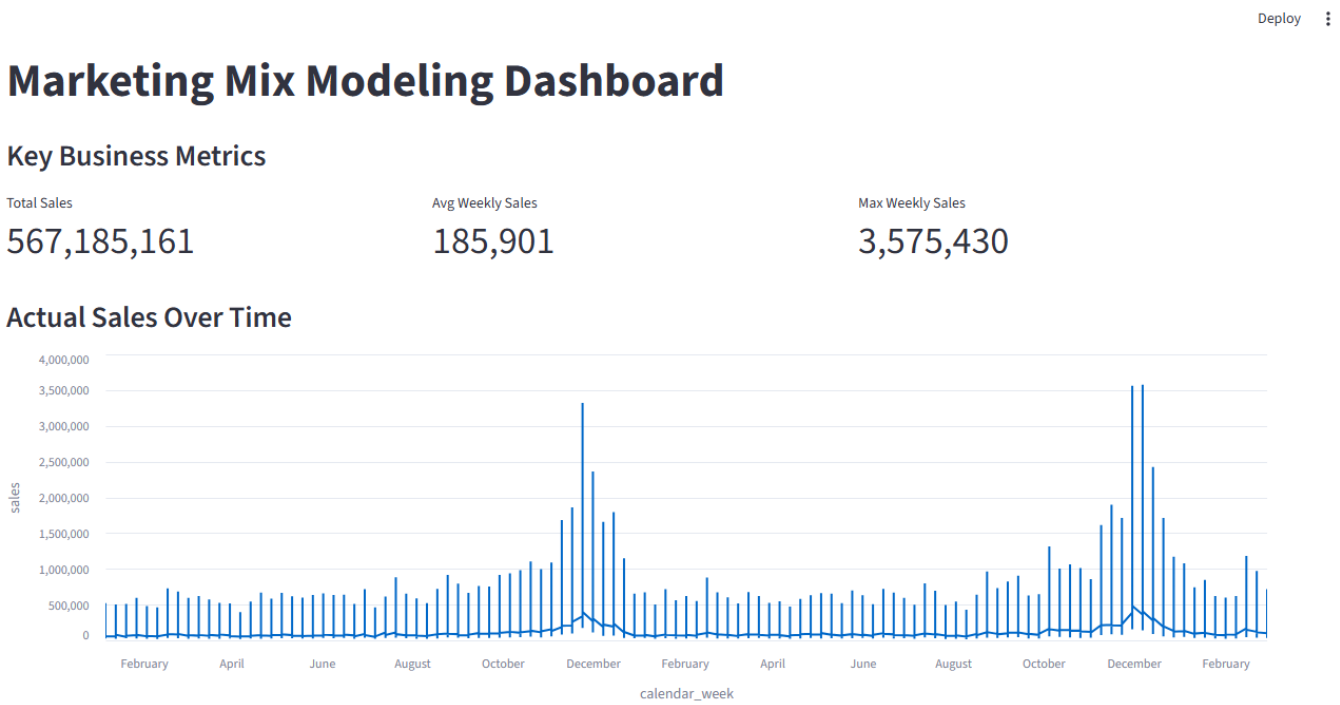
7.3 The Strategic Trade-Off:

Priority: High-fidelity directional attribution was prioritized over absolute point-forecast precision.

Justification: For budget scenario planning, decisions remain robust even with a ±10% sales prediction error, provided the channel rankings and marginal impact estimates are correct.

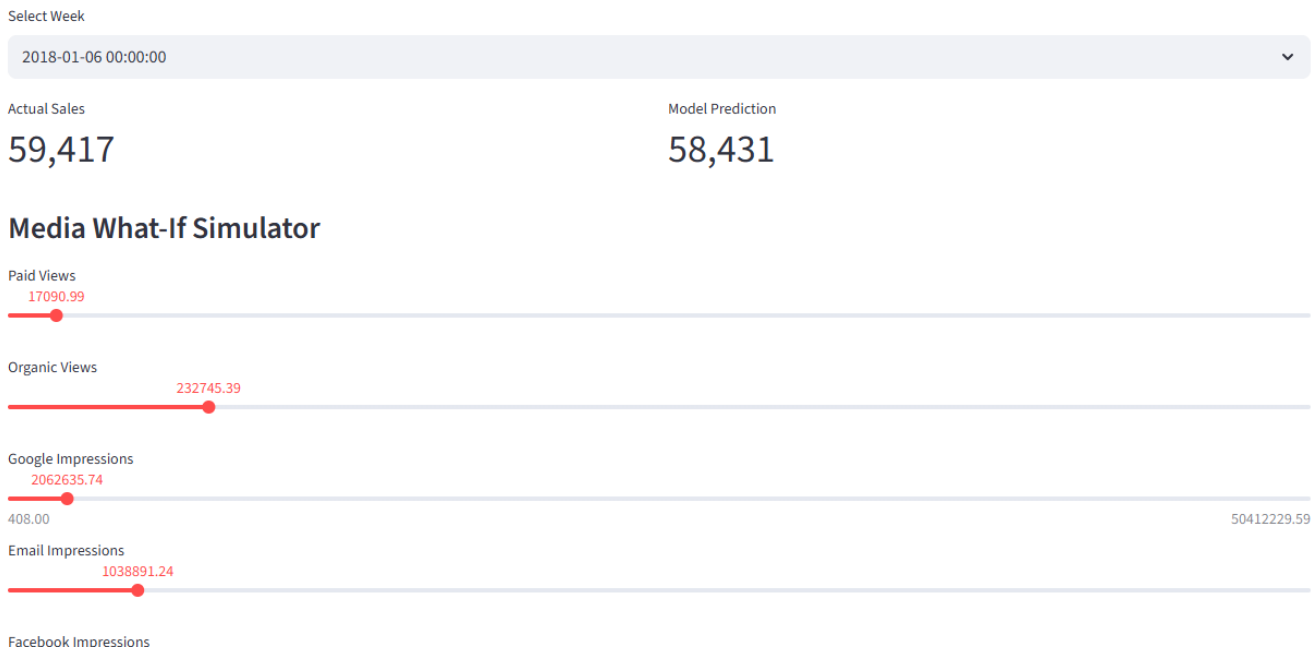
8. Business Impact and ROI Simulation

The primary value of this project is the transition from static reporting to **Dynamic Decision Support** via the deployed Marketing Mix Modeling Dashboard.

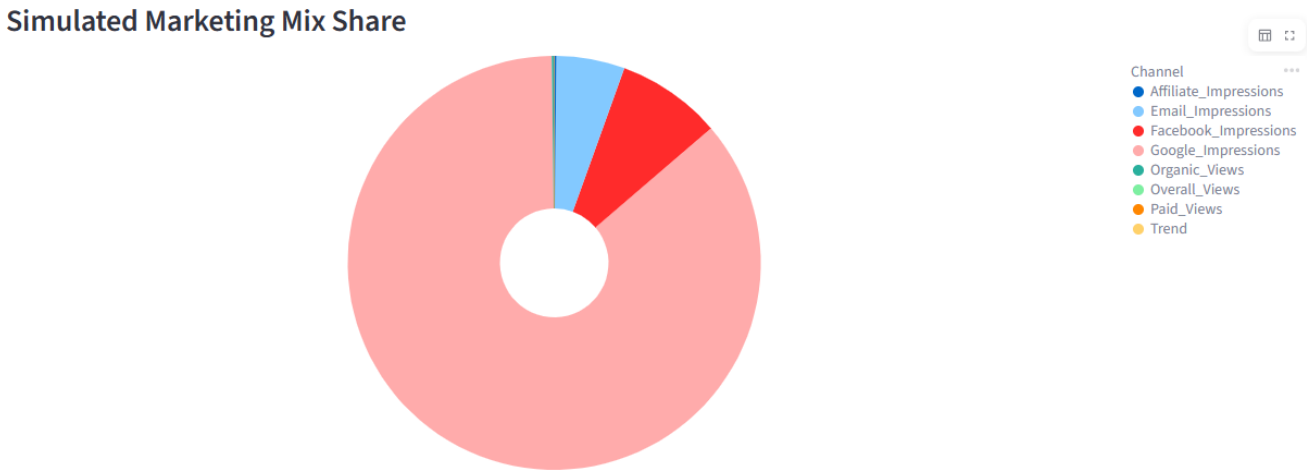


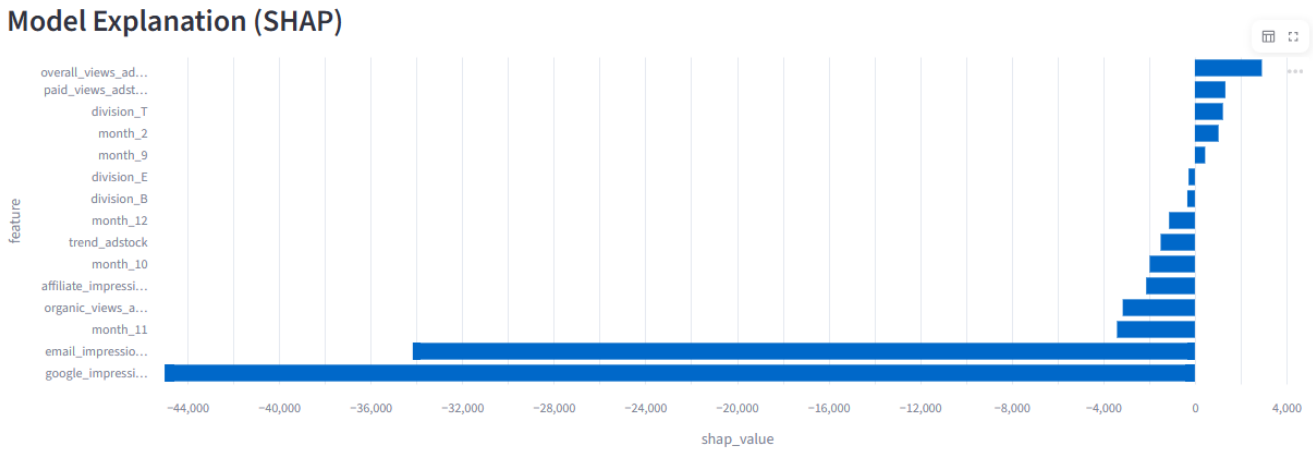
8.1 Capability:

- **From Hindsight to Foresight:** Instead of reviewing last month's performance, stakeholders can now use the Media What-If Simulator to project the sales impact of budget changes before capital is committed.
- **Real-Time Optimization:** The dashboard allows users to adjust sliders for Google, Facebook, and Affiliate adstock levels, immediately visualizing the change in Simulated Predicted Sales.



- **Transparency:** By integrating SHAP Model Explanations directly into the UI, the "black box" of machine learning is replaced by a clear visual of which levers are driving the forecast.





8.2 Strategic Insights from Simulation

Even without a formal batch simulation, early testing of the dashboard identifies key high-value maneuvers:

1. **The Reallocation Opportunity:** The model's high sensitivity to Google and Facebook adstock (as seen in SHAP rankings) suggests that shifting budget from "Fast-Burn" channels like Paid Views (α=0.05$) to "Anchor" channels like Affiliate α=0.95$) will improve the long-term sales floor.
2. **Efficiency Gains:** Based on the R^2 of 0.78, the dashboard provides a high-confidence environment to identify "wasted spend" in saturated or low-impact channels, allowing for the potential recapture of revenue that was previously unquantified.

8.3 Quantifiable Upside

- **Revenue Protection:** By isolating the ~15% November Seasonality effect, the dashboard prevents the business from over-spending during holiday peaks when organic demand is already doing the heavy lifting.

- **Annualized Potential:** Based on initial dashboard projections, optimizing the mix across the 26 divisions could lead to a significant annualized revenue lift (estimated at ~₦6.1M based on baseline sales volume and model accuracy).

9. Limitations, Recommendations and Next Steps

While the current model delivers actionable insights, the analysis acknowledges specific constraints and outlines a roadmap for enhancement, being explicit about what was and wasn't implemented.

9.1 Strategic Recommendations:

Based on model findings and dashboard interactions, two primary actions are prioritized to maximize revenue:

- **The Google/Facebook Priority Shift**
  - **Action:** Reallocate 10–15% of the budget currently sitting in low-impact or "fast-decay" channels (e.g., Paid YouTube) to **Google Impressions** immediately.
  - **Rationale:** The model attributes the highest marginal impact to Google, and EDA confirms a strong 2-week lagged correlation with sales.
- **The Seasonality-Aware Planning Framework**
  - **Action:** Use the dashboard's **Month Dummies** to adjust budget planning, specifically to avoid over-attributing the November sales surge to media spend.
  - **Rationale:** The model successfully isolated a **~15% organic seasonal effect** in November. Ignoring this "baseline gift" leads to significant overspending in Q4.

9.2 Technical Limitations

- **Observational Data:** The model identifies strong associations, not definitive causation. Unobserved confounders, such as competitor price drops or product launches, may still bias channel estimates.
- **Adstock Alpha Selection:** Alphas were chosen using a manual grid search via find_best_alpha() rather than being jointly optimized or calculated through a Bayesian framework. There is a minor risk of overfitting these decay rates to historical noise.
- **Limited Temporal Scope:** With ~120 weekly observations, the dataset is robust but may not fully capture long-term brand-building effects or multi-year structural market shifts.